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Wearable electronic device receiving information from external wearable electronic device and method for operating the same

Abstract

According to various embodiments, a wearable electronic device may comprise a display, a communication circuit, a voice input device, and at least one processor operatively connected with the display, the communication circuit, and the voice input device. The at least one processor may be configured to obtain audio data through the voice input device, identify whether the audio data satisfies a predetermined condition, receive, from an external wearable electronic device through the communication circuit, state information based on a signal obtained from the external wearable electronic device, and control the display to display visual information corresponding to the audio data, based on at least part of the state information. Other various embodiments are possible as well.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/KR2022/000998 designating the United States, filed on Jan. 19, 2022, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent

Application No. 10-2021-0008797, filed on Jan. 21, 2021 in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

1. Field

(1) The disclosure relates to a wearable electronic device receiving information from an external wearable electronic device and a method for operating the same.

2. Description of Related Art

(2) Augmented reality (AR) is technology for overlaying three-dimensional (3D) (or two-dimensional (2D)) virtual images on a real-world image or background and displaying them as overlaid images. AR technology which combines the real-world environment with virtual objects enables the user to view the real-world environment, thereby providing a better real-life feel and additional information. The user may observe the image together with the real-world environment and identify information about a target object in the environment that he is currently viewing.

(3) An augmented reality device may be a wearable electronic device. For example, AR glasses-type electronic devices, which may be worn on the face like glasses, are in wide use.

(4) Speech to text (STT) is a technology for receiving a voice, converting the input voice into a text form, and outputting it. When a wearable electronic device is worn, voices generated in the ambient environment may be less perceivable. Thus, it is possible to visually provide information about a voice generated in the ambient environment to the user of the wearable electronic device through the STT function.

(5) To provide an STT function according to a user's need, a wearable electronic device supporting the STT function may determine in what situation it is to support the STT function. For example, the wearable electronic device may operate while being worn on the user's body portion. Thus, it may be limited to obtain sufficient data to determine whether the user is under the situation where the STT function is needed, only with the wearable electronic device providing the STT function.

(6) Further, in general, a wearable electronic device may have a small size to be worn on the user's body portion and may thus obtain only data in a local environment. Thus, the accuracy of the STT function may be limited.

SUMMARY

(7) According to an embodiment, a wearable electronic device may receive, from an external wearable electronic device, state information based on a signal obtained from the external wearable electronic device and provide an STT function considering the state information.

(8) According to an embodiment, a wearable electronic device may comprise a display, a communication circuit, a voice input device, and at least one processor. The at least one processor may be configured to obtain audio data through the voice input device, identify whether the audio data satisfies a predetermined condition, receive, from an external wearable electronic device through the communication circuit, state information based on a signal obtained from the external wearable electronic device, and control the display to display visual information corresponding to the audio data, based on at least part of the state information.

(9) According to an embodiment, a wearable electronic device may comprise a display, a communication circuit, a voice input device, and at least one processor. The at least one processor may be configured to obtain first audio data corresponding to an external event through the voice input device, receive, from an external wearable electronic device through the communication circuit, second audio data corresponding to the external event and obtained from the external wearable electronic device, identify a direction corresponding to the external event, based on the first audio data and the second audio data, and perform an operation corresponding to the identified direction.

(10) According to an embodiment, a method performed in a wearable electronic device may comprise obtaining audio data, identifying whether the audio data satisfies a predetermined

condition, receiving, from an external wearable electronic device, state information based on a signal obtained from the external wearable electronic device, and displaying visual information corresponding to the audio data, based on at least part of the state information.
(11) According to an embodiment, there may be provided a wearable electronic device receiving information from an external wearable electronic device and a method for operating the same. According to an embodiment, a wearable electronic device may receive, from an external wearable electronic device, state information based on a signal obtained from the external wearable electronic device and provide an STT function considering the state information. According to an embodiment, a wearable electronic device may determine whether to provide an STT function considering state information based on a signal obtained from an external wearable electronic device and may thus accurately determine whether the user is under a situation where the STT function is needed. Further, according to an embodiment, a wearable electronic device provides an STT function based on state information which is based on a signal obtained from an external wearable electronic device. Thus, the wearable electronic device may provide an STT function with high accuracy.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates a structure of a wearable electronic device according to various embodiments;
(2) FIG. **2** illustrates a structure of a display and an eye tracking camera of a wearable electronic device according to various embodiments;
(3) FIG. **3** is a block diagram illustrating a wearable electronic device according to various embodiments;
(4) FIG. **4** is a block diagram illustrating an external wearable electronic device according to various embodiments;
(5) FIG. **5** illustrates communication between a wearable electronic device and an external wearable electronic device according to various embodiments;
(6) FIG. **6** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments;
(7) FIG. **7** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments;
(8) FIG. **8** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments;
(9) FIG. **9** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments;
(10) FIG. **10** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments;
(11) FIG. **11** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments;
(12) FIG. **12** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments;
(13) FIG. **13** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments; and
(14) FIG. **14** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments.
DETAILED DESCRIPTION
(15) As is traditional in the field, embodiments may be described and illustrated in terms of blocks which carry out a described function or functions. These blocks, as shown in the drawings, which

may be referred to herein as units or modules or the like, or by names such as device or the like, may be physically implemented by analog or digital circuits such as logic gates, integrated circuits, microprocessors, microcontrollers, memory circuits, passive electronic components, active electronic components, optical components, hardwired circuits, or the like, and may be driven by firmware and software. The circuits may, for example, be embodied in one or more semiconductor chips, or on substrate supports such as printed circuit boards and the like. Circuits included in a block may be implemented by dedicated hardware, or by a processor (e.g., one or more programmed microprocessors and associated circuitry), or by a combination of dedicated hardware to perform some functions of the block and a processor to perform other functions of the block. Each block of the embodiments may be physically separated into two or more interacting and discrete blocks. Likewise, the blocks of the embodiments may be physically combined into more complex blocks.

(16) FIG. **1** illustrates a structure of a wearable electronic device according to various embodiments. According to various embodiments, a wearable electronic device **100** may include a frame **105**, a first support portion **101**, a second support portion **102**, a first hinge portion **103** connecting the frame **105** and the first support portion **101**, and a second hinge portion **104** for connecting the frame **105** and the second support portion **102**. According to an embodiment, the frame **105** may include at least one camera, for example a first camera **111-1**, a first camera **111-2**, a second camera **112-1**, a second camera **112-2** and/or a third camera **113**, one or more light emitting elements, for example a light emitting element **114-1** and a light emitting element **114-2**, at least one display, for example a first display **151** and a second display **152**, one or more audio input devices, for example an audio input device **162-1**, an audio input device **162-2**, and an audio input device **162-3**, and one or more transparent members, for example a transparent member **190-1** and a transparent member **190-2**. According to various embodiments, the wearable electronic device **100** may include one or more first cameras **111-1** and **111-2**, one or more second cameras **112-1** and **112-2**, and one or more third cameras **113**. According to various embodiments, images obtained through one or more first cameras **111-1** and **111-2** may be used for detection of a hand gesture by the user, tracking the user's head, and spatial recognition. According to various embodiments, the one or more first cameras **111-1** and **111-1** may be global shutter (GS) cameras. According to various embodiments, the one or more first cameras **111-1** and **111-2** may perform a simultaneous localization and mapping (SLAM) operation through depth capture. According to various embodiments, the one or more first cameras **111-1** and **111-2** may perform spatial recognition for 6 degrees of freedom (DoF).

(17) According to various embodiments, an image obtained through the one or more second cameras **112-1** and **112-2** may be used to detect and track the user's pupil. According to various embodiments, the one or more second cameras **112-1** and **112-2** may be GS cameras. According to various embodiments, the one or more second cameras **112-1** and **112-2** may correspond to the left eye and the right eye, respectively, and the one or more second cameras **112-1** and **112-2** may have the same performance.

(18) According to various embodiments, the one or more third cameras **113** may be high-resolution cameras. According to various embodiments, the one or more third cameras **113** may perform an auto-focusing (AF) function and an image stabilization function. According to various embodiments, the one or more third cameras **113** may be a GS camera or a rolling shutter (RS) camera.

(19) According to various embodiments, the wearable electronic device **100** may include one or more light emitting elements **114-1** and **114-2**. The light emitting elements **114-1** and **114-2** may be different from a light source, which is described below, for irradiating light to a screen output area of the display. According to various embodiments, the light emitting elements **114-1** and **114-2** may irradiate light to facilitate pupil detection in detecting and tracking the user's pupils through the one or more second cameras **112-1** and **112-2**. According to various embodiments, each of the light

emitting elements **114-1** and **114-2** may include an LED. According to various embodiments, the light emitting elements **114-1** and **114-2** may irradiate light in an infrared band. According to various embodiments, the light emitting elements **114-1** and **114-2** may be attached around the frame **105** of the wearable electronic device **100**. According to various embodiments, the light emitting elements **114-1** and **114-2** may be positioned around the one or more first cameras **111-1** and **111-2** and may assist in gesture detection, head tracking, and/or spatial recognition by the one or more first cameras **111-1** and **111-2** when the wearable electronic device **100** is used in a dark environment. According to various embodiments, the light emitting elements **114-1** and **114-2** may be positioned around the one or more third cameras **113** and may assist in obtaining images by the one or more third cameras **113** when the wearable electronic device **100** is used in a dark environment.

(20) According to various embodiments, the wearable electronic device **100** may include a first display **151**, a second display **152**, one or more input optical members **153-1** and **153-2**, one or more transparent members **190-1** and **190-2**, and one or more screen display portions **154-1** and **154-2**, which are positioned in the frame **105**. According to various embodiments, the first display **151** and the second display **152** may include, e.g., a liquid crystal display (LCD), a digital mirror device (DMD), a liquid crystal on silicon (LCoS), or an organic light emitting diode (OLED), or a micro light emitting diode (micro LED). According to various embodiments, when the first display **151** and the second display **152** are formed of one of a liquid crystal display device, a digital mirror display device, or a silicon liquid crystal display device, the wearable electronic device may include a light source for irradiating light to a screen output area of the display. According to various embodiments, when the first display **151** and the second display **152** may generate light on their own, e.g., when formed of either organic light emitting diodes or micro LEDs, the wearable electronic device **100** may provide a virtual image of good quality to the user even when a separate light source is not included.

(21) According to various embodiments, the one or more transparent members **190-1** and **190-2** may be disposed to face the user's eyes when the user wears the wearable electronic device **100**. According to various embodiments, the one or more transparent members **190-1** and **190-2** may include at least one of a glass plate, a plastic plate, and a polymer. According to various embodiments, the user may view the outside world through the one or more transparent members **190-1** and **190-2** when the user wears the wearable electronic device **100**. According to various embodiments, the one or more input optical members **153-1** and **153-2** may guide the light generated by the first display **151** and the second display **152** to the user's eyes. According to various embodiments, images based on the light generated by the first display **151** and the second display **152** may be formed on one or more screen display portions **154-1** and **154-2** on the one or more transparent members **190-1** and **190-2**, and the user may view the images formed on the one or more screen display portions **154-1** and **154-2**.

(22) According to various embodiments, the wearable electronic device **100** may include one or more optical waveguides. The optical waveguide may transfer the light generated by the first display **151** and the second display **152** to the user's eyes. The wearable electronic device **100** may include one optical waveguide corresponding to each of the left eye and the right eye. According to various embodiments, the optical waveguide may include at least one of glass, plastic, or polymer. According to various embodiments, the optical waveguide may include a nano-pattern formed inside or on one outer surface, e.g., a polygonal or curved grating structure. According to various embodiments, the optical waveguide may include a free-form type prism, and in this case, the optical waveguide may provide incident light to the user through a reflective mirror. According to various embodiments, the optical waveguide may include at least one of at least one diffractive element, for example a diffractive optical element (DOE) or a holographic optical element (HOE), or a reflective element, for example a reflective mirror, and guide the display light emitted from the light source to the user's eyes using at least one diffractive element or reflective element included in

the optical waveguide. According to various embodiments, the diffractive element may include input/output optical elements. According to various embodiments, the reflective element may include a member causing total reflection.

(23) According to various embodiments, the wearable electronic device **100** may include the one or more audio input devices **162-1**, **162-2**, and **162-3**, and the one or more audio input devices **162-1**, **162-2**, and **162-3** may receive the user's voice or sounds generated around the wearable electronic device **100**. For example, the one or more audio input devices **162-1**, **162-2**, and **162-3** may receive the sound generated around and transfer the sound to the processor (e.g., the processor **320** of FIG. **3**) so that the wearable electronic device **100** may provide a speech-to-text (STT) function.

(24) According to various embodiments, the one or more support portions (e.g., the first support portion **101** and the second support portion **102**) may include at least one printed circuit board (PCB), for example a first PCB **170-1** and a second PCB **170-2**, one or more audio output devices, for example an audio output device **163-1** and an audio output device **163-2**, and one or more batteries, for example a battery **135-1** and a battery **135-2**. The first PCB **170-1** and the second PCB **170-2** may transfer electrical signals to components included in the wearable electronic device **100**, such as a first camera **211**, a second camera **212**, a third camera **213**, a display module **250**, an audio module **261**, and a sensor **280** described below with reference to FIG. **2**. According to various embodiments, at least one of the first PCB **170-1** and the second PCB **170-2** may be a flexible printed circuit board (FPCB). According to various embodiments, the first PCB **170-1** and the second PCB **170-2** each may include a first substrate, a second substrate, and an interposer disposed between the first substrate and the second substrate. According to various embodiments, the wearable electronic device **100** may include the batteries **135-1** and **135-2**. The batteries **135-1** and **135-2** may store power for operating the remaining components of the wearable electronic device **100**. According to various embodiments, the one or more audio output devices **163-1** and **163-2** may output audio data to the user. For example, feedback on the user's command (or input) may be provided, or information about a virtual object may be provided to the user through audio data.

(25) According to various embodiments, the wearable electronic device **100** may include one or more hinge portions (e.g., the first hinge portion **103** and the second hinge portion **104**). For example, the first hinge portion **103** may allow the first support portion **101** to be coupled to the frame **105** and rotate about the frame **105**, and the second hinge portion **104** may allow the second support portion **102** to be coupled to the frame **105** and rotate about the frame **105**.

(26) FIG. **2** illustrates a structure of a display and an eye tracking camera of a wearable electronic device according to various embodiments. A wearable electronic device **200** (e.g., the wearable electronic device **100** of FIG. **1**) may include a display **221**, an input optical member **222**, a display optical waveguide **223**, an output optical member **224**, an eye tracking camera **210**, a first splitter **241**, an eye tracking optical waveguide **242**, and a second splitter **243**.

(27) In the wearable electronic device, the display **221** may correspond to the first display **151** or the second display **152** illustrated in FIG. **1**. The light output from the display **221** may pass through the input optical member **222**, which may correspond to the input optical members **153-1** and **153-2** of FIG. **1**, and be incident on the display optical waveguide **223**, and then may pass through the display optical waveguide **223** and be output through the output optical member **224**. The light output from the output optical member **224** may be seen by the user's eyes **230**. Hereinafter, in embodiments, the expression “displays an object on the display” may mean that light output from the display **221** may be output through the output optical member **224**, and the shape of the object is seen by the user's eyes **230** by the light output through the output optical member **224**. Further, in embodiments, the expression “controls the display to display the object” may mean that the light output from the display **221** may be output through the output optical member **224**, and the display **221** is controlled so that the shape of the object is seen by the user's eyes **230** by the light output through the output optical member **224**.

(28) The light **235** reflected from the user's eye **230** may pass through the first splitter **241** and be incident on the eye tracking optical waveguide **242**, and may then pass through the eye tracking optical waveguide **242** and be output to the eye tracking camera **210** through the second splitter **243**. According to various embodiments, the light **235** reflected from the user's eye **230** may correspond to light output from the light emitting elements **114-1** and **114-2** of FIG. **1** and reflected from the user's eye **230**. According to various embodiments, the eye tracking camera **210** may correspond to the one or more second cameras **112-1** and **112-1** illustrated in FIG. **2**.
(29) According to various embodiments, a wearable electronic device **300** may include a first camera **311**, a second camera **312**, a third camera **313**, a processor **320**, a power management integrated circuit (PMIC) **330**, a battery **335**, a memory **340**, a display **350**, an audio interface **361**, an audio input device **362**, an audio output device **363**, a communication circuit **370**, and a sensor **380**.
(30) According to various embodiments, the details of the one or more first cameras **111-1** and **111-2**, one or more second cameras **112-1** and **112-2**, and one or more third cameras **113** described above in connection with FIG. **1** may be equally applied to the first camera **311**, the second camera **312**, and the third camera **313**, respectively. According to various embodiments, the wearable electronic device **300** may include at least one of the first camera **311**, the second camera **312**, and the third camera **313**, in plurality.
(31) According to various embodiments, the processor **320** may control other components of the wearable electronic device **300**, e.g., the first camera **311**, the second camera **312**, the third camera **313**, the PMIC **330**, the memory **340**, the display **350**, the audio interface **361**, the communication circuit **370**, and the sensor **380** and may perform various data processing or computations.
(32) According to various embodiments, the PMIC **330** may convert the power stored in the battery **335** to have the current or voltage required by the other components of the wearable electronic device **300** and supply power to other components of the wearable electronic device **300**.
(33) According to various embodiments, the memory **340** may store various data used by at least one component (e.g., the processor **320** or a sensor **380**) of the wearable electronic device **300**.
(34) According to various embodiments, the display **350** may display a screen to be provided to the user. According to various embodiments, the display **350** may include the first display **151**, the second display **152**, one or more input optical members **153-1** and **153-2**, one or more transparent members **190-1** and **190-2**, and one or more screen display portions **154-1** and **154-2**, which are described above in connection with FIG. **1**.
(35) According to various embodiments, the audio interface **361** may be connected to the audio input device **362** and the audio output device **363** and may convert the data input through the audio input device **362** and may convert data to be output to the audio output device **363**. In embodiments, the audio input device **362** may include a microphone, and the audio output device **363** may include a speaker and an amplifier.
(36) According to various embodiments, the communication circuit **370** may support establishment of a wireless communication channel with an electronic device outside the wearable electronic device **300** and performing communication through the established communication channel.
(37) According to various embodiments, the sensor **380** may include a 6-axis sensor **381**, a magnetic sensor **382**, a proximity sensor **383**, and an optical sensor **384**. According to various embodiments, the sensor **380** may include a sensor for obtaining a biometric signal for detecting whether the wearable electronic device **300** is being worn by the user. For example, the sensor **380** may include at least one of a heart rate sensor, a skin sensor, and a temperature sensor.
(38) According to various embodiments, when the user activates the STT function, the processor **320** may generate the text- and/or image-based data to be displayed through the display **350** based on the data received from the audio interface **361**.
(39) According to various embodiments, the processor **320** may identify the movement of the user wearing the wearable electronic device **300** through the 6-axis sensor **381**. For example, the 6-axis

sensor **381** may generate a sensor value by detecting a change in the direction the user faces (e.g., the direction the user views through the wearable electronic device **300**) and may transfer the generated sensor value or a variation in the sensor value to the processor **320**.
(40) According to various embodiments, when the user activates the STT function, the audio interface **361** may receive a sound generated around the wearable electronic device **300** (or the user) through the audio input device **362** and may transfer the data obtained by converting the received sound to the processor **320**.
(41) According to various embodiments, the communication circuit **370** may transmit and receive data to and from an external electronic device (e.g., a wearable electronic device such as an earphone, or an external electronic device such as a terminal). For example, the wearable electronic device **300** may receive audio data received by the external wearable electronic device through the communication circuit **370** and may transfer the received audio data to the processor **320**. As another example, the wearable electronic device **300** may output image data, which is based on data received from the external electronic device through the communication circuit **370**, through the display **350**.
(42) FIG. **4** is a block diagram illustrating an external wearable electronic device **400** according to various embodiments. According to various embodiments, an external wearable electronic device **400** may be at least one of an earphone-type wearable electronic device, a watch-type wearable electronic device, or a necklace-type wearable electronic device. According to various embodiments, the external wearable electronic device **400** may have a plurality of physically separated housings. For example, when the external wearable electronic device **400** is an earphone-type wearable electronic device, the external wearable electronic device **400** may include a first housing to be worn on the left ear and a second housing to be worn on the right ear. In embodiments, the components illustrated in FIG. **4** may be included in one or more of the plurality of housings.
(43) According to various embodiments, the external wearable electronic device **400** may include a processor **410**, a memory **420**, a communication circuit **430**, an audio interface **440**, a sensor **450**, and a battery **460**.
(44) According to various embodiments, the processor **410** may receive data from other components of the external wearable electronic device **400**, e.g., the memory **420**, the communication circuit **430**, the audio interface **440**, the sensor **450**, and the battery **460**, perform computation based on the received data, and transfer signals to control the other components to the other components. According to various embodiments, the processor **410** may operate based on instructions stored in the memory **420**.
(45) According to various embodiments, the memory **420** may store instructions to enable other components of the external wearable electronic device **400**, e.g., the processor **410**, the communication circuit **430**, the audio interface **440**, the sensor **450**, and the battery **460**, to perform designated operations. According to various embodiments, the memory **420** may store audio data obtained through the audio interface **440**.
(46) According to various embodiments, the communication circuit **430** may perform wireless communication with another electronic device (e.g., the wearable electronic device **300**). According to various embodiments, the communication circuit **430** may transmit information obtained from the external wearable electronic device **400** to the wearable electronic device **300**. The type of communication supported by the communication circuit **430** is not limited.
(47) According to various embodiments, the audio interface **440** may include a plurality of microphones and one or more speakers. According to various embodiments, the plurality of microphones may include a microphone that faces toward the user's inner ear when the user wears the external wearable electronic device **400** and a microphone that faces away from the user when the user wears the external wearable electronic device **400**. According to various embodiments, the audio interface **440** may obtain audio data through each of the plurality of microphones and may

perform noise cancellation based on the audio data obtained through the plurality of microphones.
(48) According to various embodiments, the sensor **450** may include a biometric sensor for detecting whether the user wears the external wearable electronic device **400**. For example, the biometric sensor may include at least one of a heart rate sensor, a skin sensor, and a temperature sensor. According to various embodiments, the sensor **450** may include a geomagnetic sensor.
(49) According to various embodiments, the external wearable electronic device **400** may receive a data transmission request from the wearable electronic device **300** through the communication circuit **430**. For example, the external wearable electronic device **400** may receive a request to transmit the audio data received through the audio interface **440**. According to an embodiment, when a designated condition (e.g., detection of a designated motion or a designated time) occurs from the wearable electronic device **300**, the external wearable electronic device **400** may receive a request to transmit the audio data received through the audio module **440**.
(50) FIG. **5** illustrates communication between a wearable electronic device and an external wearable electronic device according to various embodiments. FIG. **5** illustrates communication between a wearable electronic device **510**, an external wearable electronic device (R) **520**, and an external wearable electronic device (L) **530** in a case where the external wearable electronic device is an earphone-type wearable electronic device and includes the external wearable electronic device (L) **530** to be worn on the left ear and the external wearable electronic device (R) **520** to be worn on the right ear.
(51) According to various embodiments, one of the external wearable electronic device (R) **520** and the external wearable electronic device (L) **530** may operate as a master, and the other may act as a slave. FIG. **5** illustrates an example in which the external wearable electronic device (R) **520** operates as a master, and the external wearable electronic device (L) **530** operates as a slave.
(52) In FIG. **5**, the wearable electronic device **510** and the external wearable electronic device (R) **520** may be connected to each other through the Bluetooth communication protocol, and the external wearable electronic device (R) **520** and the external wearable electronic device (L) **530** may be connected to each other through the Bluetooth communication protocol. According to various embodiments, the external wearable electronic device (R) **520** may perform communication with the wearable electronic device **510**. The external wearable electronic device (L) **530** may receive information about the communication link between the wearable electronic device **510** and the external wearable electronic device (R) **520** from the external wearable electronic device (R) **520**. According to various embodiments, the information about the communication link between the wearable electronic device **510** and the external wearable electronic device (R) **520** may include address information, clock information, channel information, session description protocol (SDP) result information, information about supported functions, key information, or an extended inquiry response (EIR) packet. The external wearable electronic device (L) **530** may monitor the communication channel between the wearable electronic device **510** and the external wearable electronic device (R) **520** based on the information about the communication link between the wearable electronic device **510** and the external wearable electronic device (R) **520**. For example, the external wearable electronic device (L) **530** may receive the data transmitted/received through the communication channel between the wearable electronic device **510** and/or the external wearable electronic device (R) **520** by the wearable electronic device **510** and the external wearable electronic device (R) **520**. As another example, the external wearable electronic device (L) **530** may transmit data to the wearable electronic device **510** through the communication channel between the wearable electronic device **510** and the external wearable electronic device (R) **520**.
(53) The wearable electronic device **510** may send a request for state information about the external wearable electronic device R **520** and the external wearable electronic device L **530**, obtained by the external wearable electronic device R **520** and the external wearable electronic device L **530**, to the external wearable electronic device R **520**. Details of examples of the state information are described below with reference to FIG. **6**. The wearable electronic device **510** may receive state

information from the external wearable electronic device (R) **520**. The state information obtained from the external wearable electronic device (L) **530** may be transmitted to the wearable electronic device **510** during retransmission periods between the external wearable electronic device (R) **520** and the wearable electronic device **510**, indicated by W**1** and W**2** in FIG. **5**.

(54) FIG. **6** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments. According to an embodiment, when the user wears the wearable electronic device, the wearable electronic device (e.g., the wearable electronic device **300**) may receive audio data generated around and may receive a request to activate the STT function which provides text and/or images based on the received audio data. According to another embodiment, when the user wears the wearable electronic device, the wearable electronic device **300** may automatically activate the STT function. According to another embodiment, when the user wears the wearable electronic device, the wearable electronic device **300** may determine whether the user is outputting audio data through another wearable electronic device (e.g., an earphone) and may activate the STT function based on the result of the determination.

(55) According to an embodiment, in operation **610**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may obtain audio data. According to various embodiments, the processor **320** may obtain audio data through the audio interface **361** of the wearable electronic device **300**.

(56) In operation **620**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the audio data obtained in operation **610** meets a predetermined condition. According to various embodiments, the predetermined condition may be met when the audio data indicates a situation in which the user of the wearable electronic device **300** may want to receive the STT function. For example, the predetermined condition may include at least one of when the audio data includes a language-related voice, when the audio data includes a preset word-related voice, or when the audio data includes a voice having a preset volume or higher. In embodiments, preset may mean, for example, predetermined, or determined at a previous time.

(57) In operation **620**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the obtained audio data meets the predetermined condition. In one example, when it is identified that the obtained audio data does not meet the predetermined condition in operation **620**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may repeat operation **610** until audio data meeting the predetermined condition is obtained.

(58) In another example, if it is identified that the obtained audio data meets the predetermined condition in operation **620**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may receive state information from the external wearable electronic device (e.g., the external wearable electronic device **400**) through the communication circuit **370** in operation **630**. According to various embodiments, the state information may be information based on a signal obtained from the external wearable electronic device **400**.

(59) According to various embodiments, the state information may indicate whether the external wearable electronic device **400** is being worn by the user of the external wearable electronic device **400**. According to various embodiments, the state information indicating whether the external wearable electronic device **400** is being worn by the user may be a biometric signal obtained from the sensor **450** of the external wearable electronic device **400** or information output based on the biometric signal obtained by the sensor **450**, from the processor **410** of the external wearable electronic device **400** and may indicate the result of determining whether the external wearable electronic device **400** is being worn by the user.

(60) According to various embodiments, the state information may indicate whether a voice is being output from the external wearable electronic device **400**. According to various embodiments, the state information may indicate the volume of the voice being output from the external wearable

electronic device **400**.

(61) According to various embodiments, in a case where the external wearable electronic device **400** provides a noise canceling function and ambient sound listening function and allows the user to set priority on the noise canceling function and the ambient sound listening function, the state information may indicate which one of the noise canceling function and the ambient sound listening function has a higher priority.

(62) According to various embodiments, the state information may indicate whether the ambient sound listening function is active in the external wearable electronic device **400** by the user's input to the external wearable electronic device **400**.

(63) According to various embodiments, the state information may include audio data obtained through the audio interface **440** of the external wearable electronic device **400**.

(64) In operation **640**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display (e.g., the display **350**) to display visual information corresponding to the audio data based at least part of the state information. According to various embodiments, the visual information may include at least one of text or an image.

(65) According to various embodiments, that the processor **320** controlling the display **350** to display visual information corresponding to audio data based at least part of the state information may mean that the processor **320** determines whether to provide the STT function based on at least part of the state information.

(66) According to various embodiments, when the state information indicates that the external wearable electronic device **400** is being worn by the user of the external wearable electronic device **400**, that the state information indicates that the external wearable electronic device **400** is being worn may be included in at least one condition for the processor **320** to control the display **350** to display the visual information corresponding to the audio data. In embodiments, the at least one condition for the processor **320** to control the display **350** to display the visual information corresponding to the audio data may be referred to as “at least one visual information display condition”. In other words, the processor **320** may further have any of various conditions to be described below as conditions for providing the STT function in addition to a condition that the external wearable electronic device **400** is being worn. According to various embodiments, when the state information indicates that the external wearable electronic device **400** is not being worn, the processor **320** may not control the display **350** to display visual information corresponding to the audio data.

(67) According to various embodiments, when the state information is a biometric signal obtained from the sensor **450** of the external wearable electronic device **400**, the processor **320** may obtain a biometric signal through the sensor **380** of the wearable electronic device **300** and compare the biometric signal received from the external wearable electronic device **400** with the biometric signal obtained through the sensor **380** of the wearable electronic device **300**, thereby identifying whether the user wearing the external wearable electronic device **400** is identical to the user wearing the wearable electronic device **300**. According to various embodiments, the at least one visual information display condition may include a condition that the state information indicates that the external wearable electronic device **400** is being worn, and also a condition that that the user wearing the external wearable electronic device **400** is identical to the user wearing the wearable electronic device **300**. According to various embodiments, when it is identified that the user wearing the external wearable electronic device **400** is not identical to the user wearing the wearable electronic device **300**, the processor **320** may not control the display **350** to display the visual information corresponding to the audio data.

(68) According to various embodiments, when the state information indicates whether the external wearable electronic device **400** is outputting a voice, the at least one visual information display condition may include a condition that the state information indicates that the external wearable

electronic device **400** is outputting a voice.
(69) According to various embodiments, when the state information indicates the volume of the voice being output from the external wearable electronic device **400**, the at least one visual information display condition may include a condition that the volume of the voice being output from the external wearable electronic device **400**, indicated by the state information, is a preset level or more.
(70) According to various embodiments, when the state information indicates relative priorities between the noise canceling function and the ambient sound listening function, the at least one visual information display condition may include a condition that the priority of the noise canceling function indicated by the state information is higher than the priority of the ambient sound listening function.
(71) According to various embodiments, when the state information indicates whether the ambient sound listening function is active in the external wearable electronic device **400** by the user's input to the external wearable electronic device **400**, the at least one visual information display condition may include a condition that the ambient sound listening function is not activated by the user's input to the external wearable electronic device **400**. In other words, when the user activates the ambient sound listening function through a direct input to the external wearable electronic device **400**, the wearable electronic device **300** may not provide the STT function.
(72) According to various embodiments, the processor **320** controlling the display **350** to display visual information corresponding to audio data based at least part of the state information may mean that the processor **320** considers the state information in determining the visual information to be provided, when the processor **320** provides the STT function.
(73) According to various embodiments, when the state information includes audio data obtained through the audio interface **440** of the external wearable electronic device **400**, the processor **320** may process the audio data obtained in operation **610** based on the audio data obtained through the audio interface **440** of the external wearable electronic device **400**, thereby obtaining third audio data and controlling the display **350** to display visual information corresponding to the third audio data. According to various embodiments, the processing for obtaining the third audio data may be noise canceling processing for removing ambient noise except for voice conversations.
(74) According to various embodiments, when providing the STT function, the processor **320** may adjust the visual feature of the visual information displayed on the display **350** according to the volume of the voice corresponding to the audio data obtained through the audio interface **440** of the external wearable electronic device **400**, obtained as at least part of the state information and/or the audio data obtained in operation **610**. For example, when the visual information is text, at least one of the font, size, or color of the text may be adjusted according to the volume of the voice corresponding to the audio data. As another example, when the visual information is an image, at least one of the size or color of the image may be adjusted according to the volume of the voice corresponding to the audio data.
(75) FIG. **7** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments. According to an embodiment, a wearable electronic device (e.g., the wearable electronic device **300**) may perform communication connection with an external wearable electronic device (e.g., the external wearable electronic device **400**). For example, when the wearable electronic device **300** and the external wearable electronic device **400** are positioned in a short distance, the wearable electronic device **300** may perform communication connection (e.g., Bluetooth communication connection) with the external wearable electronic device through a communication circuit (e.g., the communication circuit **370**).
(76) According to an embodiment, in operation **710**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may receive state information from the external wearable electronic device (e.g., the external wearable electronic device **400**). According to various embodiments, the state information may indicate whether the external

wearable electronic device **400** is being worn by the user and whether a voice is being output from the external wearable electronic device **400**.
(77) In operation **720**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the external wearable electronic device **400** is being worn by the user based on the state information. In one example, when it is identified that the external wearable electronic device **400** is not being worn by the user, the method may be terminated.
(78) When it is identified that the external wearable electronic device **400** is being worn by the user in operation **720**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether a voice is being output from the external wearable electronic device **400** based on the state information in operation **730**. In an example, when it is identified that the external wearable electronic device **400** is not outputting a voice, the method may be terminated.
(79) When it is identified that the external wearable electronic device **400** is outputting a voice in operation **730**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display **385** to display visual information corresponding to the audio data obtained through the audio interface **361** of the wearable electronic device **300** in operation **740**. According to various embodiments, the visual information may include at least one of text or an image.
(80) In embodiments, the order of operation **720** and operation **730** may be changed. For example, operation **730** may be performed before operation **720**.
(81) In embodiments, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may perform operations **610** and **620** of FIG. **6** before performing operation **710** and, in response to identifying that the audio data meets a predetermined condition in operation **620**, perform operation **710**. In other words, in addition to a condition that the external wearable electronic device **400** is being worn, and a condition that the external wearable electronic device **400** is outputting voice, conditions for providing the STT service, for example the at least one visual information display condition, may also include a condition related to the audio data.
(82) FIG. **8** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments. In operation **810**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may receive state information from the external wearable electronic device (e.g., the external wearable electronic device **400**). In operation **820**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the external wearable electronic device **400** is being worn by the user based on the state information. In operation **830**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the external wearable electronic device **400** is outputting a voice based on the state information. The details of operations **710**, **720**, and **730** described above in connection with FIG. **7** may be applied likewise to operations **810**, **820**, and **830**.
(83) When it is identified that the external wearable electronic device **400** is outputting a voice in operation **830**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display **385** to display a visual indicator indicating that the STT service may be provided in operation **840**. For example, the visual indicator may be a virtual object including text and/or image indicating that there is information to be provided to the user, based on the audio interface **440** included in the external wearable electronic device **400** and/or the audio input device **362** included in the wearable electronic device **300**.
(84) In operation **850**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may determine whether a reaction condition regarding the user of the wearable electronic device **300** is met. According to various embodiments, the reaction

condition may be a condition related to the reaction of the user of the wearable electronic device **300** to the visual indicator displayed in operation **840**. According to various embodiments, the processor **320** may identify the user's gaze through the second camera **312** and, if it is identified that the user's gaze is on the visual indicator for a preset first time or longer, may identify that the reaction condition is met. According to various embodiments, the processor **320** may analyze the user's utterance and, when a preset utterance for receiving the STT service is detected, may identify that the reaction condition is met. In this case, the processor **320** may store data related to the user's voice in the memory **340** and may identify whether the preset utterance is the user's utterance based on the stored user's voice data. According to various embodiments, the processor **320** may detect a gesture through the first camera **311** and, when a preset gesture for receiving the STT service is detected, may identify that the reaction condition is met. According to various embodiments, if any combination of the above-described example reaction conditions is met, the processor **320** may identify that the reaction condition is met.
(85) In operation **850**, when it is identified that the reaction condition is not met, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may repeat operation **850** until the reaction condition is met.
(86) When it is identified that the reaction condition is met in operation **850**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display **385** to display visual information corresponding to the audio data obtained through the audio interface **361** of the wearable electronic device **300** in operation **860** According to various embodiments, the visual information may include at least one of text or an image.
(87) According to various embodiments, the processor **320** may store audio data gathered before the reaction condition is met in the memory **340** and, if it is identified that the reaction condition is met, control the display **385** to further display the visual information corresponding to the audio data gathered before the reaction condition is met, as well as the visual information corresponding to the audio data gathered after the reaction condition is met in operation **860**.
(88) In embodiments, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may perform operations **610** and **620** of FIG. **6** before performing operation **810** and, in response to identifying that the audio data meets a predetermined condition in operation **620**, perform operation **810**.
(89) FIG. **9** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments. In operation **910**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may receive state information from the external wearable electronic device (e.g., the external wearable electronic device **400**). According to various embodiments, the state information may indicate which function of the noise canceling function and the ambient sound listening function is set to have a higher priority in the external wearable electronic device **400**.
(90) In operation **920**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the priority of the noise canceling function of the external wearable electronic device **400** is higher than the priority of the ambient sound listening function based on the state information. For example, the priority of the ambient sound listening function of the external wearable electronic device **400** being higher than the priority of the noise canceling function of the external wearable electronic device **400** may mean that the user desires to listen to ambient sound through the external wearable electronic device **400** rather than receiving the STT function based on the external sound through the wearable electronic device **300**.
(91) According to an embodiment, the external wearable electronic device **400** may detect that the user activates the ambient sound listening function through detection of a designated motion or the user's utterance, or the user's input, such as a touch, tap, or long-press, through the sensor **450**.
(92) When it is determined that the priority of the noise canceling function of the external wearable

electronic device **400** is higher than the priority of the ambient sound listening function in operation **920**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display **385** to display visual information corresponding to the audio data obtained through the audio interface **361** of the wearable electronic device **300** in operation **930**. According to various embodiments, the visual information may include at least one of text or an image.
(93) If it is determined that the priority of the noise canceling function of the external wearable electronic device **400** is lower than the priority of the ambient sound listening function in operation **920**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may terminate the method.
(94) In embodiments, when it is determined that the priority of the noise canceling function of the external wearable electronic device **400** is higher than the priority of the ambient sound listening function in operation **920**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may determine whether the condition for providing the STT function is met, as described below with reference to FIG. **12** and, when the condition of FIG. **12** is met, control the display **385** to display visual information corresponding to the audio data obtained through the audio interface **361** of the wearable electronic device **300** in operation **12**.
(95) In embodiments, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may perform operations **610** and **620** of FIG. **6** before performing operation **910** and, in response to identifying that the audio data meets a predetermined condition in operation **620**, perform operation **910**.
(96) FIG. **10** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments. In operation **1010**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may obtain first audio data. According to various embodiments, the processor **320** may obtain first audio data through the audio interface **361** of the wearable electronic device **300**.
(97) In operation **1020**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the first audio data obtained in operation **1010** meets a predetermined condition. Details of the predetermined condition may be the same as those described above with reference to operation **620** of FIG. **6**.
(98) When it is identified that the obtained first audio data does not meet the predetermined condition in operation **1020**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may repeat operations **1010** and **1020** until first audio data meeting the predetermined condition is obtained.
(99) In operation **1020**, when it is identified that the first audio data meets the predetermined condition, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may obtain state information including second audio data obtained through the audio interface **440** of the external wearable electronic device **400**, from the external wearable electronic device (e.g., the external wearable electronic device **400**) in operation **1030**. For example, the wearable electronic device **300** may transmit data including information (e.g., reception time information and/or sampling data) about the first audio data to the external wearable electronic device **400** and receive state information including the second audio data based on the information about the first audio data from the external wearable electronic device **400**. According to an embodiment, before determining whether the first audio data meets the predetermined condition, the wearable electronic device **300** may perform communication connection with the external wearable electronic device **400** through the communication circuit **370** and obtain the state information including the second audio data from the external wearable electronic device **400**.
(100) In operation **1040**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may obtain third audio data by processing the first audio data based on the second audio data. According to various embodiments, in operation **1040**, the

processor **320** may perform noise canceling processing to remove ambient noise except for the voice conversations from the first audio data.
(101) In operation **1050**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display **350** to display the visual information corresponding to the third audio data obtained in operation **1040**. According to various embodiments, the visual information may include at least one of text or an image.
(102) In embodiments, the processor **320** may receive state information indicating whether the external wearable electronic device **400** is being worn by the user and whether the external wearable electronic device **400** is outputting a voice in operation **1030** and, without immediately performing operation **1040** after performing operation **1030**, perform operations **720** and **730** of FIG. **7** and, when the conditions of operations **720** and **730** are met, perform operation **1040**.
(103) Further, according to various embodiments, the processor **320** may receive state information indicating whether the external wearable electronic device **400** is being worn by the user and whether the external wearable electronic device **400** is outputting a voice in operation **1030** and, without immediately performing operation **1040** after performing operation **1030**, perform operations **820** to **850** of FIG. **8** and, when the conditions of operations **820**, **830**, and **850** are met, perform operation **1040**.
(104) FIG. **11** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments. In operation **1110**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display **350** to display visual information corresponding to the audio data obtained through the audio interface **361** of the wearable electronic device **300**. According to various embodiments, the visual information may include at least one of text or an image.
(105) In operation **1120**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether a condition for stopping the providing of visual information is met. According to various embodiments, the condition for stopping providing visual information may be a condition indicating that it is appropriate to stop providing the STT service. According to various embodiments, the condition for stopping providing visual information may include a condition that the user's gaze is not on the visual information displayed in operation **1110** for greater than or equal to a preset time. According to various embodiments, the condition for stopping providing visual information may include a condition of detecting a preset gesture to request to stop the STT service by the user. According to various embodiments, the condition for stopping providing visual information may include a condition that the accuracy of sentences included in the visual information displayed in operation **1110** is less than or equal to a preset level. For example, the accuracy of sentences included in the visual information may be determined based on the completeness of the sentences and/or the accuracy of the context. According to various embodiments, the condition for stopping providing visual information may include a condition that the visual information displayed in operation **1110** is displayed for greater than a designated time.
(106) In operation **1120**, when it is identified that the condition for stopping providing visual information is met in operation **1120**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display **350** not to display the visual information corresponding to the audio data in operation **1130**.
(107) When it is identified that the condition for stopping providing visual information is not met in operation **1120**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may repeat operation **1120** until the condition for stopping providing visual information is identified to be met while controlling the display **350** to continue to display the visual information corresponding to the audio data obtained through the audio interface **361**.
(108) In embodiments, when the condition for stopping providing visual information includes the

condition that the accuracy of sentences included in the visual information displayed in operation **1110** is less than or equal to a preset level, and the accuracy of sentences included in the visual information is identified to be less than or equal to the preset level in operation **1120**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may transmit a signal for activating the ambient sound listening function to the external wearable electronic device **400** through the communication circuit **370**.

(109) FIG. **12** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments. In operation **1210**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may obtain first audio data corresponding to an external event through the audio input device **362** of the wearable electronic device **300**. According to various embodiments, the external event may include an utterance by a person other than the user of the wearable electronic device **300**. For example, the external event may include generation of a sound corresponding to a designated condition for example a sound having a signal strength larger than or equal to a designated signal strength, from the outside, for example an outside of any of wearable electronic devices **100**, **200**, or **300**.

(110) In operation **1220**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may receive second audio data corresponding to the external event and obtained from the external wearable electronic device **400**, from the external wearable electronic device (e.g., the external wearable electronic device **400**) through the communication circuit **370**.

(111) In operation **1230**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify the direction corresponding to the external event, based on the first audio data and the second audio data. According to an embodiment, the processor **320** of the wearable electronic device **300** may determine the direction corresponding to the external event based on the position of at least one audio interface **440** of the external wearable electronic device **400** and the position of at least one audio input device **362** of the wearable electronic device **300**. For example, the processor **320** may determine the direction corresponding to the external event based on time information about reception of the first audio data and time information about reception of the second audio data. According to various embodiments, the direction corresponding to the external event may be a relative direction of the position where the external event occurs, relative to the wearable electronic device **300**.

(112) In operation **1240**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may perform the operation corresponding to the identified direction.

(113) According to various embodiments, the processor **320** may identify the gaze direction, of the user of the wearable electronic device **300**, or the direction in which the user views through the transparent member (e.g., one or more transparent members **190-1** and **190-2**) of the wearable electronic device **300**, based on the data obtained through the sensor **380** and/or at least one camera (e.g., the first camera **311** or the second camera **312**) of the wearable electronic device **300** and perform different operations depending on whether the identified user's gaze direction is identical to the direction identified in operation **1230**. According to various embodiments, the processor **320** may identify the gaze direction of the user of the wearable electronic device **300** based on data obtained through the second camera **312**. According to various embodiments, the processor **320** may identify the direction in which the wearable electronic device **300** faces based on the data obtained through the sensor **380** of the wearable electronic device **300** and identify the identified direction as the user's gaze direction. According to various embodiments, when the user's gaze direction is identical to the direction corresponding to the external event identified in operation **1230**, the processor **320** may transmit a signal for activating the ambient sound listening function to the external wearable electronic device **400** through the communication circuit **370**. According to various embodiments, when the user's gaze direction is not identical to the direction identified in

operation **1230**, the processor **320** may control the display **350** to display the visual information corresponding to the external event based on at least one of the first audio data or the second audio data.
(114) FIG. **13** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments. In operation **1310**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may obtain first audio data corresponding to an external event through the audio input device **362** of the wearable electronic device **300**.
(115) In operation **1320**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the first audio data meets a predetermined condition. Details of the predetermined condition may be the same as those described above with reference to operation **620** of FIG. **6**.
(116) When it is identified that the obtained first audio data does not meet the predetermined condition in operation **1320**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may repeat operations **1310** and **1320** until first audio data meeting the predetermined condition is obtained.
(117) If it is identified that the first audio data meets the predetermined condition in operation **1320**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may receive state information from the external wearable electronic device (e.g., the external wearable electronic device **400**). According to various embodiments, the state information may indicate whether the external wearable electronic device **400** is being worn by the user and whether a voice is being output from the external wearable electronic device **400**, and the state information may include the second audio data obtained through the audio interface **440** of the external wearable electronic device **400**. According to an embodiment, the wearable electronic device **300** may perform a communication connection with the external wearable electronic device **400** through the communication circuit **370** before determining whether the first audio data meets a predetermined condition. For example, the communication may include short-range communication, such as Bluetooth or Wi-Fi.
(118) In operation **1340**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the external wearable electronic device **400** is being worn and whether the external wearable electronic device **400** is outputting a voice based on the state information. According to various embodiments, as described above in connection with operation **640** of FIG. **6**, in operation **1340**, the processor **320** may further identify whether the user wearing the external wearable electronic device **400** is identical to the user wearing the wearable electronic device **300** and, when it is identified that the user wearing the external wearable electronic device **400** is identical to the user wearing the wearable electronic device **300**, perform operation **1350**. According to an embodiment, in operation **1340**, the processor **320** may terminate the method when the external wearable electronic device **400** is not worn or is not outputting a voice.
(119) When it is identified in operation **1340** that the external wearable electronic device **400** is worn and the external wearable electronic device **400** is outputting a voice, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may receive the second audio data obtained through the audio interface **440** of the external wearable electronic device **400** from the external wearable electronic device (e.g., the external wearable electronic device **400**) through the communication circuit **370** in operation **1350**.
(120) In operation **1360**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify the direction corresponding to the external event, based on the first audio data and the second audio data. The details of operation **1230** may be likewise applied to operation **1360**.
(121) In operation **1370**, the processor (e.g., the processor **320**) of the wearable electronic device

(e.g., the wearable electronic device **300**) may identify the user's gaze direction, or the direction in which the user views through the transparent member (e.g., one or more transparent members **190-1** and **190-2** of FIG. **1**) of the wearable electronic device **300**, based on the data obtained through at least one of the sensor **380** or the second camera **312**. According to various embodiments, the processor **320** may identify the gaze direction of the user of the wearable electronic device **300** based on data obtained through the second camera **312**. According to various embodiments, the processor **320** may identify the direction in which the wearable electronic device **300** faces based on the data obtained through the sensor **380** of the wearable electronic device **300** and identify the identified direction as the user's gaze direction.
(122) In operation **1380**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the direction corresponding to the external event is identical to the user's gaze direction.
(123) When it is identified in operation **1380** that the direction corresponding to the external event is identical to the user's gaze direction, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may transmit a signal for activating the ambient sound listening function to the external wearable electronic device **400** through the communication circuit **370** in operation **1390**.
(124) When it is identified in operation **1380** that the direction corresponding to the external event is not identical to the user's gaze direction, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display **350** to display the visual information corresponding to the external event based on at least one of the first audio data or the second audio data in operation **1395**. According to various embodiments, the visual information may include at least one of text or an image.
(125) FIG. **14** is a flowchart illustrating operations performed in a wearable electronic device according to various embodiments. According to an embodiment, the user wears the wearable electronic device **300** and the external wearable electronic device **400** and configures it to activate the STT function and display visual information corresponding to the external event. In operation **1410**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display **350** to display the visual information corresponding to the external event. For example, the processor **320** may display, on the display **350**, visual information based on the third audio data which is based on the first audio data received from the wearable electronic device **300** and the second audio data received from the external wearable electronic device **400**. According to various embodiments, the visual information may include at least one of text or an image.
(126) In operation **1420**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify the user's second gaze direction. A process of identifying the user's second gaze direction may be identical to operation **1370** of FIG. **13**. Here, the term ‘second gaze direction’ may mean the user's gaze direction while the visual information corresponding to the external event is displayed, i.e., while the STT function is provided. The term ‘second gaze direction’ may be different from the gaze direction in operation **1370** of FIG. **13**, that is, the user's gaze direction before the STT function is provided, which may be for example a first gaze direction.
(127) In operation **1430**, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may identify whether the user's second gaze direction is identical to the direction corresponding to the external event. According to various embodiments, similar to operation **1360** of FIG. **13**, the processor **320** may identify the direction corresponding to the external event based on the second audio data received from the external wearable electronic device **400** and the first audio data obtained through the audio input device **362**.
(128) When it is identified in operation **1430** that the user's second gaze direction is not identical to the direction corresponding to the external event, the processor (e.g., the processor **320**) of the

wearable electronic device (e.g., the wearable electronic device **300**) may continuously provide the STT function while repeating operations **1410** to **1430** until it is identified that the user's second gaze direction is identical to the direction corresponding to the external event.
(129) When it is identified in operation **1430** that the user's second gaze direction is identical to the direction corresponding to the external event, the processor (e.g., the processor **320**) of the wearable electronic device (e.g., the wearable electronic device **300**) may control the display **350** to stop displaying the visual information corresponding to the external event and transmit a signal for activating the ambient sound listening function to the external wearable electronic device **400** through the communication circuit **370** in operation **1440**.
(130) According to various embodiments, a wearable electronic device (e.g., the wearable electronic device **300**) may include a display (e.g., the display **350**), a communication circuit (e.g., the communication circuit **370**), a voice input device (e.g., the audio input device **362**), and a processor (e.g., the processor **320**) operatively connected with the display, the communication circuit, and the voice input device. The processor **320** may be configured to obtain audio data through the audio input device **362**, identify whether the audio data satisfies a predetermined condition, receive, from an external wearable electronic device (e.g., the external wearable electronic device **400**) through the communication circuit **370**, state information based on a signal obtained from the external wearable electronic device **400**, and control the display **350** to display visual information corresponding to the audio data, based on at least part of the state information.
(131) According to various embodiments, the predetermined condition may include at least one of a condition in which the audio data includes a language-related voice, a condition in which the audio data includes a preset word-related voice, or a condition in which the audio data includes a voice having a volume greater than or equal to a preset volume.
(132) According to various embodiments, the state information may indicate whether the external wearable electronic device **400** is being worn by a user.
(133) According to various embodiments, the state information may include first data obtained from a first biometric sensor (e.g., the sensor **380**) of the external wearable electronic device **400**. The wearable electronic device **300** may include a second biometric sensor (e.g., the sensor **380**). The processor **320** may be configured to obtain second data through the second biometric sensor, identify that the user wearing the external wearable electronic device is wearing the wearable electronic device **400**, based on the first data and the second data, and control the display **350** to display the visual information corresponding to the audio data, based on identifying that the user wearing the external wearable electronic device **400** is wearing the wearable electronic device **300**.
(134) According to various embodiments, the state information may indicate whether the external wearable electronic device **400** is outputting a voice. The processor **320** may be configured to control the display to display **350** the visual information corresponding to the audio data, based on identifying that the external wearable electronic device **400** is being worn by the user and the external wearable electronic device **400** is outputting the voice.
(135) According to various embodiments, the processor **320** may be configured to control the display **350** to display a visual indicator indicating that a speech to text (STT) service is providable, based on identifying that the external wearable electronic device **400** is being worn by the user, and the external wearable electronic device **400** is outputting the voice, and control the display **350** to display the visual information corresponding to the audio data, in response to a reaction condition regarding the user being satisfied while the visual indicator is displayed on the display **350**.
(136) According to various embodiments, the reaction condition may include at least one of a condition in which the user's gaze is on the visual indicator for a time greater than or equal to a preset first time, a condition in which a preset utterance by the user is detected, or a condition in which a preset first gesture by the user is detected.
(137) According to various embodiments, the processor **320** may be configured to control the display **350** to further display the visual information corresponding to the audio data before the

reaction condition is satisfied.
(138) According to various embodiments, the state information may indicate that a priority of a noise canceling function of the external wearable electronic device **400** is higher than a priority of an ambient sound listening function of the external wearable electronic device **400**. The processor **320** may be configured to control the display **350** to display the visual information corresponding to the audio data, based on identifying that the priority of the noise canceling function of the external wearable electronic device **400** is higher than the priority of the ambient sound listening function of the external wearable electronic device **400**.
(139) According to various embodiments, the state information may include second audio data obtained from the external wearable electronic device **400**.
(140) According to various embodiments, the processor **320** may be configured to obtain third audio data by processing the audio data based on the second audio data and control the display **350** to display visual information corresponding to the third audio data.
(141) According to various embodiments, the processor **320** may be configured to adjust a visual feature of the visual information displayed on the display **350** according to a volume of a voice corresponding to the audio data.
(142) According to various embodiments, the processor **320** may be configured to identify whether a condition for stopping providing the visual information is satisfied while controlling the display **350** to display the visual information corresponding to the audio data and control the display **350** not to display the visual information corresponding to the audio data, based on the condition for stopping providing the visual information being satisfied. The condition for stopping providing the visual information may include at least one of a condition in which a time during which the user's gaze is not on the visual information displayed on the display **350** lasts for a time greater than or equal to a preset second time, a condition in which a preset second gesture by the user is detected, or a condition in which an accuracy of sentences included in the visual information corresponding to the audio data is a level less than or equal to a preset level.
(143) According to various embodiments, the wearable electronic device **300** may be communication-connected with an external electronic device (e.g., a smartphone) through the communication circuit **370**, and the external wearable electronic device **400** may be communication-connected with an external electronic device through the communication circuit **430**.
(144) According to an embodiment, the wearable electronic device **300** may transmit, to the external electronic device through the communication circuit **370**, the data received through at least one camera (e.g., the first camera **111-1** or **111-2**, the second camera **112-1** or **112-2**, and/or the third camera **113**) or one or more audio input devices **162-1**, **162-2**, and **162-3**. According to an embodiment, the wearable electronic device **300** may output visual information through at least one display (e.g., the first display **151**, the second display **152**, or the display **350**) or output a voice through at least one audio output device **363**, based on the data received from the external electronic device.
(145) According to an embodiment, the external electronic device may obtain audio data from the wearable electronic device **300** and/or the external wearable electronic device **400** and provide an STT function based on the obtained audio data. According to an embodiment, the external electronic device may include at least one or more audio input device. The external electronic device may obtain audio data corresponding to an external event through the audio input device. When the audio data corresponding to the external event meets a designated condition, the external electronic device may request the wearable electronic device **300** and/or the external wearable electronic device **400** to transmit audio data. The external electronic device may obtain audio data from the wearable electronic device **300** and/or the external wearable electronic device **400** and provide an STT function based on the obtained audio data.
(146) According to an embodiment, the external electronic device may receive first audio data

through the wearable electronic device **300** and receive second audio data through the external wearable electronic device **400**. According to an embodiment, the external electronic device may generate third audio data based on the first audio data and the second audio data and transmit visual information, based on the generated third audio data, to the wearable electronic device **300** to be output.

(147) According to an embodiment, the external electronic device may receive a redirection of the user's gaze to a direction corresponding to the external event, from the external wearable electronic device **400** and/or the wearable electronic device **300** while outputting the visual information based on the third audio data through the wearable electronic device **300**. For example, the wearable electronic device **300** may transmit, to the external electronic device, the data obtained from at least one camera (e.g., the first camera **111-1** or **111-2**, the second camera **112-1** or **112-2**, and/or the third camera **113**) and/or the sensor **380**. As another example, the external wearable electronic device **400** may transmit the data obtained from the sensor **450** to the external electronic device.

(148) According to an embodiment, the external electronic device may identify a redirection of the user's gaze to the direction corresponding to the external event based on the data received from the external wearable electronic device **400** and/or the wearable electronic device **300**. According to an embodiment, when the user's gaze is redirected to the direction corresponding to the external event, the external electronic device may request the wearable electronic device **300** to stop outputting the visual information through the display **350** and request the external wearable electronic device **400** to activate the ambient sound listening function.

(149) According to various embodiments, a wearable electronic device (e.g., the wearable electronic device **300**) may include a display (e.g., the display **350**), a communication circuit (e.g., the communication circuit **370**), a voice input device (e.g., the audio input device **362**), and a processor (e.g., the processor **320**) operatively connected with the display **350**, the communication circuit **370**, and the audio input device **362**. The processor **320** may be configured to obtain first audio data corresponding to an external event through the audio input device **362**, receive, from an external wearable electronic device (e.g., the external wearable electronic device **400**) through the communication circuit **370**, second audio data corresponding to the external event and obtained from the external wearable electronic device **400**, identify a direction corresponding to the external event, based on the first audio data and the second audio data, and perform an operation corresponding to the identified direction.

(150) According to various embodiments, the processor **320** may be configured to receive second audio data based on the first audio data satisfying a predetermined first condition. The predetermined condition may include at least one of a condition in which the first audio data includes a language-related voice, a condition in which the first audio data includes a preset word-related voice, or a condition in which the first audio data includes a voice having a volume greater than or equal to a preset volume.

(151) According to various embodiments, the processor **320** may be configured to receive state information from the external wearable electronic device **400** through the communication circuit **370** and receive the second audio data, based on the state information indicating that the external wearable electronic device **400** is being worn and outputting a voice.

(152) According to various embodiments, the wearable electronic device **300** may further include a sensor module (e.g., the sensor **380**). The processor **320** may be configured to identify a gaze direction of a user of the wearable electronic device **300** through the sensor **380**, identify whether a direction corresponding to the external event is identical to the user's gaze direction, and transmit a signal for activating an ambient sound listening function to the external wearable electronic device **400**, through the communication circuit **370**, based on the direction corresponding to the external event being identical to the user's gaze direction.

(153) According to various embodiments, the processor **320** may be configured to control the display **350** to, based on the direction corresponding to the external event being not identical to the

user's gaze direction, display, visual information corresponding to the external event, based on at least one of the first audio data or the second audio data.

(154) According to various embodiments, the processor **320** may be configured to identify the user's second gaze direction through the sensor module while the visual information is displayed on the display **350**, identify that the direction corresponding to the external event is identical to the second gaze direction, and based on the direction corresponding to the external event being identical to the second gaze direction, control the display **350** to stop displaying the visual information corresponding to the external event and transmit a signal for activating the ambient sound listening function to the external wearable electronic device **400**, through the communication circuit **370**.

(155) According to various embodiments, a method performed in a wearable electronic device (e.g., the wearable electronic device **300**) may include obtaining audio data, identifying whether the audio data satisfies a predetermined condition, receiving, from an external wearable electronic device (e.g., the external wearable electronic device **400**), state information based on a signal obtained from the external wearable electronic device **400**, and displaying visual information corresponding to the audio data, based on at least part of the state information.

(156) It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(157) As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(158) Various embodiments as set forth herein may be implemented as software (e.g., the program **340**) including one or more instructions that are stored in a storage medium (e.g., internal memory **336** or external memory **338**) that is readable by a machine (e.g., the electronic device **301**). For example, a processor (e.g., the processor **320**) of the machine (e.g., the electronic device **301**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(159) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program products may be traded as commodities between sellers and buyers. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., Play Store™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.
(160) According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. Some of the plurality of entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. A wearable electronic device comprising: a display; communication circuitry; a voice input device; at least one processor; and memory configured to store instructions which, when executed by the at least one processor, cause the wearable electronic device to: obtain audio data through the voice input device, identify whether the audio data satisfies a predetermined condition, receive, from an external wearable electronic device through the communication circuitry, state information based on a signal obtained from the external wearable electronic device, wherein the state information indicates whether the external wearable electronic device is outputting a voice, and based on identifying that the external wearable electronic device is outputting the voice: determine to perform speech to text converting, convert an input voice in the audio data into text form, and control the display to display visual information corresponding to the input voice in the text form, based on at least part of the state information.
2. The wearable electronic device of claim 1, wherein the predetermined condition includes at least one of a condition that the audio data includes a language-related voice, a condition that the audio data includes a predetermined word-related voice, or a condition that the audio data includes a voice having a volume greater than or equal to a predetermined volume.
3. The wearable electronic device of claim 1, wherein the state information indicates whether the external wearable electronic device is being worn by a user.
4. The wearable electronic device of claim 3, wherein the state information includes first data obtained from a first biometric sensor of the external wearable electronic device, wherein the wearable electronic device includes a second biometric sensor, and wherein the instructions, when executed by the at least one processor, cause the wearable electronic device to: obtain second data through the second biometric sensor, identify that the user wearing the external wearable electronic device is wearing the wearable electronic device, based on the first data and the second data, and control the display to display the visual information corresponding to the audio data, based on identifying that the user wearing the external wearable electronic device is wearing the wearable electronic device.

5. The wearable electronic device of claim 3, wherein the instructions, when executed by the at least one processor, cause the wearable electronic device to: control the display to display a visual indicator indicating that a speech to text service is available, based on identifying that the external wearable electronic device is being worn by the user, and the external wearable electronic device is outputting the voice, and control the display to display the visual information corresponding to the audio data, in response to a reaction condition regarding the user being satisfied while the visual indicator is displayed on the display.
6. The wearable electronic device of claim 5, wherein the reaction condition includes at least one of: a condition a gaze of the user is directed toward the visual indicator for greater than or equal to a predetermined first time, a condition that a predetermined utterance by the user is detected, or a condition that a predetermined first gesture by the user is detected.
7. The wearable electronic device of claim 5, wherein the instructions, when executed by the at least one processor, cause the wearable electronic device to control the display to further display the visual information corresponding to the audio data before the reaction condition is satisfied.
8. The wearable electronic device of claim 1, wherein the state information indicates that a priority of a noise canceling function of the external wearable electronic device is higher than a priority of an ambient sound listening function of the external wearable electronic device, and wherein the instructions, when executed by the at least one processor, cause the wearable electronic device to control the display to display the visual information corresponding to the audio data, based on identifying that the priority of the noise canceling function is higher than the priority of the ambient sound listening function.
9. The wearable electronic device of claim 1, wherein the audio data comprises first audio data, and wherein the state information includes second audio data obtained from the external wearable electronic device.
10. The wearable electronic device of claim 9, wherein the instructions, when executed by the at least one processor, cause the wearable electronic device to: obtain third audio data by processing the first audio data based on the second audio data, and control the display to display visual information corresponding to the third audio data.
11. The wearable electronic device of claim 1, wherein the instructions, when executed by the at least one processor, cause the wearable electronic device to adjust a visual feature of the visual information according to a volume of a voice corresponding to the audio data.
12. The wearable electronic device of claim 1, wherein the instructions, when executed by the at least one processor, cause the wearable electronic device to: identify whether a condition for stopping providing the visual information is satisfied while controlling the display to display the visual information corresponding to the audio data, and control the display to stop the displaying of the visual information corresponding to the audio data, based on the condition for stopping providing the visual information being satisfied, and wherein the condition for stopping providing the visual information includes at least one of: a condition a gaze of a user is not directed toward the visual information for a time greater than or equal to a predetermined second time, a condition that a predetermined second gesture by the user is detected, or a condition that an accuracy of at least one sentence included in the visual information corresponding to the audio data is less than or equal to a predetermined level.
13. A wearable electronic device comprising: a display; a communication circuit; a voice input device; a sensor; at least one processor; and a memory configured to store instructions which, when executed by the at least one processor, cause the wearable electronic device to: obtain first audio data corresponding to an external event through the voice input device, receive, from an external wearable electronic device through the communication circuit, second audio data corresponding to the external event and obtained from the external wearable electronic device, identify a direction corresponding to the external event based on the first audio data and the second audio data, identify a gaze direction of a user of the wearable electronic device using the sensor, identify whether the

direction corresponding to the external event corresponds to the gaze direction, and based on the direction corresponding to the external event corresponding to the gaze direction, control the communication circuit to transmit a signal for activating an ambient sound listening function to the external wearable electronic device.

14. The wearable electronic device of claim 13, wherein the at least one processor is further configured to receive the second audio data, based on the first audio data satisfying a predetermined first condition, and wherein the predetermined first condition includes at least one of a condition that the first audio data includes a language-related voice, a condition that the first audio data includes a predetermined word-related voice, or a condition that the first audio data includes a voice having a volume greater than or equal to a predetermined volume.

15. The wearable electronic device of claim 13, wherein the at least one processor is further configured to: receive state information from the external wearable electronic device through the communication circuit, and receive the second audio data, based on the state information indicating that the external wearable electronic device is being worn and is outputting a voice.

16. The wearable electronic device of claim 13, wherein, based on the direction corresponding to the external event being not identical to the gaze direction, the at least one processor is further configured to control the display to display, based on at least one of the first audio data or the second audio data, visual information corresponding to the external event.

17. The wearable electronic device of claim 16, wherein the gaze direction comprises a first gaze direction, and wherein the instructions further cause the wearable electronic device to: identify the a second gaze direction using the sensor while the visual information is displayed on the display, identify that the direction corresponding to the external event is identical to the second gaze direction, and based on the direction corresponding to the external event being identical to the second gaze direction, control the display to stop the displaying of the visual information corresponding to the external event, and control the communication circuit to transmit a signal for activating the ambient sound listening function to the external wearable electronic device.

18. A method performed in a wearable electronic device, the method comprising: obtaining audio data; identifying whether the audio data satisfies a predetermined condition; receiving, from an external wearable electronic device, state information based on a signal obtained from the external wearable electronic device, wherein the state information indicates whether the external wearable electronic device is outputting a voice; and based on identifying that the external wearable electronic device is outputting the voice: determining to perform speech to text converting, converting an input voice in the audio data into text form, and displaying visual information corresponding to the input voice in the text form, based on at least part of the state information.